

Attachment 5

Wastewater Pretreatment Plant (WWPP) Chemical Treatment Process Wiping Solution Recycling Plant (WSRP)

PROCESS DESCRIPTION: The used wiping solution coming from the intaglio presses is conveyed to the storage tank TK101. If maintenance on the recycling section is required, the effluent can be conveyed to the treatment tank TK403.

TK101 is connected by overflow to the emergency tanks TK100-100A, where the wiping solution can be stored in case of maintenance and then pumped to the pre-treatment section (tank TK403). TK101 flows to the process tank TK102 where the polymer Nalco 8100 is dosed to begin treatment. From TK102 the material is transferred to TK201 where perlite is added. Perlite is added by screw conveyor from TK202. The material is then transferred to either of the filter presses of the recycling system FP305 or FP405. Please note that the decanters shown in the diagram are not used. They may be used in the future for materials storage or possibly removed.

The sludge falls into containers placed beneath the filter press. Before operating in production the automatic stand-by cycle makes the effluent circulate back to the feeding tank TK201. This represents the filtration warm-up (maximum 5 minutes) which properly distributes the pressure on the plate surface and removes impurities from the cloths before starting the production.

The filtered effluent is sent to the recycled solution storage tank TK301. The recovered solution contained in the tank TK301 is pumped to the mixing tank TK315, through a set of filter cartridges, then the solution is re-heated and lost material is made up from raw materials as described earlier in the document.

The recycled wiping solution is then pumped to the printing presses by centrifugal pumps.

Spills from the system are collected by the floor sumps and returned to the system for pretreatment. For any major spill within the plant, the entire system is surrounded by a sealed curb which creates a secondary containment around the system to contain the spill.

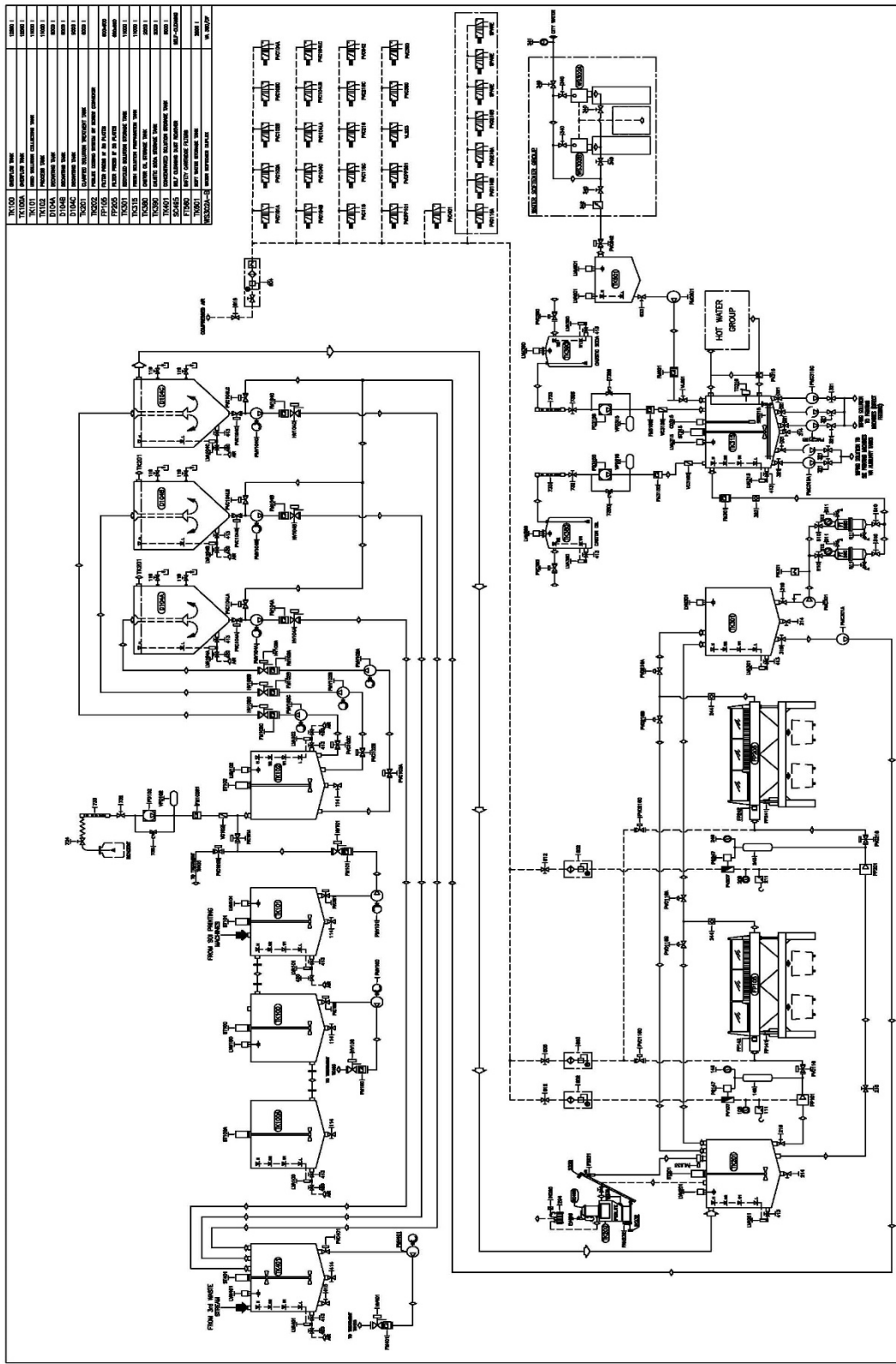
The recycled solution in tank TK315 is reconstituted by adding soft water from tank TK601 and with caustic soda solution from TK390 and castor oil from TK380 from the corresponding dosing lines by using conductivity as the control.

PROCESS OPERATION: The Contractor shall operate the system with prudent engineering and operating practices. Operating the system includes, ***but is not limited to***, the following tasks:

- (1) The Contractor shall take readings every two (2) hours of the pumps data, flows, tank levels, data set points and operating conditions from the computer, and other set points throughout the system. It should be noted that in the future this may be at least partially performed by SCADA software.
- (2) The Contractor shall perform an inspection/analysis of the WSRP every two (2) hours minimally or more often as needed for proper operation.

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- (3) The Contractor shall start, stop, empty filter presses, prepare the pallets, boxes, liners, fill each box to between $>2,350$ pounds and $\leq 2,400$ pounds, band, properly label, and enter the data into the computer logging system which is furnished by the BEP, for the performance of this contract.
- (4) The Contractor shall change/clean filter press filter cloths as needed when they become blinded or physically damaged.
- (5) The Contractor shall mix the system materials as needed for the optimal operation of the system.
- (6) The Contractor shall check and perform maintenance on control devices weekly at a minimum or more often, as needed.
- (7) The Contractor shall respond to all alarms, correcting the conditions to assure proper operation of the system.
- (8) The Contractor shall clean and inspect the tanks at least biannually as well as clean the tanks and affected equipment if there is a blockage (build-up) of sludge in any tank as necessary during the operation of the system.



New Wiping Solution Recycling - Make-up Diagram